

Math 421

Monday, November 24

Announcements

- This week's drop-in hours:
 - **Tuesday** 9-11 am VV B205
 - Wednesday 12-1 pm VV 311
- Check for any final exam conflicts & email me

Differentiable Implies Continuous

Theorem 28.2. If f is differentiable at a , then f is continuous at a .

Discussion.

Discussion Questions

Q1: Is the converse of the theorem true?

Q2: There exist functions that are *continuous everywhere* but *differentiable nowhere*. These are difficult to construct, but can you try to describe what it might look like? It could be helpful to first think of the following things:

- What might it look like for a function to be continuous but not differentiable at a *finite number of points*?
- What might it look like for a function to be continuous but not differentiable at *every integer*?

Algebraic Derivative Rules

Theorem 28.3. Let f and g be differentiable at $x = a$ and let $c \in \mathbb{R}$. Then

- i. $(cf)'(a) = cf'(a)$.
- ii. $(f + g)'(a) = f'(a) + g'(a)$.
- iii. [product rule] $(fg)'(a) = f'(a)g(a) + f(a)g'(a)$.
- iv. [quotient rule] if $g(a) \neq 0$, then $\left(\frac{f}{g}\right)'(a) = \frac{g(a)f'(a) - f(a)g'(a)}{(g(a))^2}$.

Activity

Prove parts i & ii using the limit definition of the derivative and the algebraic limit theorems.

i. $(cf)'(a) = cf'(a).$

ii. $(f + g)'(a) = f'(a) + g'(a).$

Proof of Product Rule

Theorem 28.3. (iii) Let f and g be differentiable at $x = a$ and let $c \in \mathbb{R}$. Then $(fg)'(a) = f'(a)g(a) + f(a)g'(a)$.

Discussion.

Power Rule

Theorem. For all natural numbers $n \in \mathbb{N}$, if $f(x) = x^n$, then $f'(x) = nx^{n-1}$.

Activity: Prove the power rule.

Q: What proof technique could you use?

Activity – Negative Power Rule

Let m be a positive integer, and let $h(x) = x^{-m}$ for $x \neq 0$. Then $h(x) = f(x)/g(x)$ where $f(x) = 1$ and $g(x) = x^m$. Use the quotient rule to find $h'(a)$.

Chain rule

Theorem 28.4. If f is differentiable at a and g is differentiable at $f(a)$, then $g \circ f$ is differentiable at a and

$$(g \circ f)'(a) = g'(f(a)) \cdot f'(a)$$

“Proof” of Chain Rule

Q: What is wrong with this proof?